

Genome-wide identification and expression profiling of DNA methyltransferase gene family in maize

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Abstract

Key message In this study, we identified eight DNA MTase genes in maize and the diversity of expression patterns of them was presented by EST mining, microarray and semi-quantitative expression profile analyses.

Abstract DNA methylation plays a pivotal role in promoting genomic stability through diverse biological processes including regulation of gene expression during development and chromatin organization. Although this important biological process is mainly regulated by several conserved Cytosine-5 DNA methyltransferases encoded by a smaller multigene family in plants, investigation of the plant C5-MTase-encoding gene family will serve to elucidate the epigenetic mechanism diversity in plants. Recently, genome-wide identification and evolutionary

analyses of the C5-MTase-encoding gene family have been characterized in multiple plant species including *Arabidopsis*, rice, carrot and wheat. However, little is known regarding the C5-MTase-encoding genes in the entire maize genome. Here, genome-wide identification and expression profile analyses of maize C5-MTase-encoding genes (*ZmMETs*) were performed from the latest version of the maize (B73) genome. Phylogenetic analysis indicated that the orthologs from the three species (maize, *Arabidopsis* and rice) were categorized into four classes. Chromosomal location of these genes revealed that they are unevenly distributed on 6 of all 10 chromosomes with three chromosomal/segmental duplication events, suggesting that gene duplication played a key role in expansion of the maize C5-MTase-encoding gene family. Furthermore, EST expression data mining, microarray data and semi-quantitative expression profile analyses detected in the leaves by two different abiotic stress treatments have demonstrated that these genes had temporal and spatial expression pattern and exhibited different expression levels in stress treatments, suggesting that functional diversification of *ZmMET* genes family. Overall, our study will serve to present significant insights to explore the plant C5-MTase-encoding gene expression and function and also be beneficial for future experimental research to further unravel the mechanisms of epigenetic regulation in plants.

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Introduction

With the implement of large-scale genome sequencing, transposable elements have been found that they make up a

substantial proportion of most plant genomes (Lisch 2009). Because they are potentially highly mutagenic, transposons can be usually dangerous to host genomes. Given the vast number of potentially active elements in most genomes, global activation would result in an overwhelmingly high level of mutation. This may give rise to serious consequences in plant genomes. However, it is very surprising that transposons in all species examined in detail appear to be largely quiescent, both transpositionally and transcriptionally inactive (Lisch 2009). Thus, plants could evolve a set of mechanisms whose function is to recognize and epigenetically silence them. These mechanisms amount to a genomic immune system whose function is to distinguish transposon from host. Of the genomic immune system, DNA methylation has been implied that plays an underlying role in the epigenetic regulation of transposons (Kidwell and Lisch 2001). DNA methylation has been implicated in the silencing of transposable elements and genes, and loss of methylation can have severe consequences for the organisms (Chinnusamy and Zhu 2009).

In plants, DNA methylation mainly has two roles: it protects the genome from invading and mobile DNA elements (such as transgenes, viruses, transposons, and retroelements) (Pavlopoulou and Kossida 2007) and regulates gene expression (Pósfai et al. 1989; Kumar et al. 1994). These functions have been demonstrated successfully through the endogenous genome sequences of the model plant *Arabidopsis thaliana* (Cao et al. 2003; Chen and Li 2004; Cao and Jacobsen 2002). Consistent with the primary role of DNA methylation in genome defense, most of methylated DNA sequences are composed of transposons and other dispersed repeat sequences in *A. thaliana* genome (Zhang et al. 2006). These studies revealed that DNA methylation could represent a common defense mechanism that protects the genome from those DNA elements including transposons. As shown in previous studies, transposons were mobilized when DNA methylation was reduced in *A. thaliana* (Lisch 2009; Kidwell and Lisch 2001). Moreover, in some cases, DNA methylation can control endogenous gene expression as well as the silencing of transposons. For example, the *Arabidopsis* FWA transcription factor can be regulated by DNA methylation in the developmental process of flowering (Cao and Jacobsen 2002). In other cases, DNA methylation also controls the overall level of expression from a family of repeated genes (Bartee et al. 2001). This further demonstrated that DNA methylation is essential for normal development and plays important roles in regulating gene expression in plants. However, in plants, the pathways that control the establishment and maintenance of DNA methylation as well as those involved in the removal of DNA methylation are less well characterized (Zhu 2008).

The most commonly modified base in DNA among the eukaryotes through animals and plants is 5-methylcytosine (m5C) (Lister et al. 2009; Weber and Schubeler 2007). In plants, cytosine methylation is present in not only CG and CHG trinucleotides but also in some asymmetrical context (CHH), in which H is any nucleotide (A, T or C), while in animals DNA methylation usually occurs at cytosines located in CG dinucleotides (Gruenbaum et al. 1981). Moreover, in plants the methylated patterns are inherited over multiple generations, as opposed to mammals (Kakutani 2002). Methylation of cytosine residues is enzymatically catalyzed by DNA methyltransferases (DMTs). Cytosine-5 DNA methyltransferases (C5-MTases) mainly accomplish two functions: recognition of a specific DNA sequence and catalysis of the transfer of a methyl group from the cofactor S-adenosyl-L-methionine (AdoMet) to carbon 5 in the pyrimidine ring of cytosine residues (Pósfai et al. 1989; Kumar et al. 1994). These functions are separated into two domains: the variable N-terminal domain and the catalytic C-terminal domain, respectively. The C-terminal catalytic region in plants has a structural organization similar to the bacterial restriction methyltransferases. The C-terminus contains 8 of the 10 conserved amino acid motifs diagnostic of prokaryote methyltransferases separated by variable regions (Pósfai et al. 1989; Kumar et al. 1994), suggesting that both eukaryotes and prokaryotes have retained a common reaction mechanism of methylating DNA. The catalysis is accomplished by the binding of C5-MTases to the DNA bases and the cofactor AdoMet (Jeltsch 2006). Patterns of methylation in higher plants are regulated by two distinct but complementary activities: de novo methylation and maintenance methylation (Chen and Li 2004). De novo methylation activity can methylate previously unmethylated cytosine residues, leading to the formation of new methylated patterns, whereas maintenance methylation activity mainly regulates the preexisting methylation patterns after DNA replication (Chen and Li 2004; Pavlopoulou and Kossida 2007).

In plants, genes encoding DNA methyltransferase have been identified in many species including *Arabidopsis* (Finnegan and Dennis 1993), rice (Teerawanichpan et al. 2004), carrot (Bernacchia et al. 1998) and wheat (Dai et al. 2005). Sequence analysis revealed that DNA methyltransferase genes can be classified into four distinct families, namely Dnmt1-like/MET, Dnmt2-like, CMT and Dnmt3-like/DRM, respectively (Cao et al. 2000). The Dnmt1-like/MET class acts primarily as maintenance methyltransferase, which transfers methyl groups to cytosine in hemimethylated CpG sites after DNA replication (Bestor and Verdine 1994), resulting in stable patterns of methylation that are maintained throughout development or, in many cases, between generations. The Dnmt2-like class does not appear to play a significant role in establishing or

maintaining DNA methylation patterns, and the function of these Dnmt2 methyltransferase-like proteins is unclear. The Dnmt3-like/DRM class acts as de novo methyltransferase that transfers methyl group to unmethylated DNA templates. The chromomethylases (CMTs) represent a class of DNA methyltransferases that was found only in plants and catalyze CNG methylation.

Although the expression of *ZmMET1*, a gene encoding a DNA methyltransferase from the maize MET subfamily has been detected (Steward et al. 2000), very little is known about the impact of exogenous factors on expression of DNA methyltransferase genes in maize, despite increasing knowledge on the essential roles of DNA methylation, for example in gene silencing and/or self-defense. Transposable elements have been shown to be heavily methylated, probably for silencing by the host plants, and vernalization was proposed to be associated with demethylation of genomic DNA. In an attempt to understand DNA methyltransferase roles in plants, we selected maize as an experimental material, because previous studies have relatively well characterized its genome silenced by hypermethylation and the included multi-copies of transposons and highly repetitive regions (Zhang et al. 2014; Deleris et al. 2012; Wang et al. 2009). In the present study, we analyzed the DNA methyltransferase gene family in maize, including gene structure, phylogenetic relationships, protein conserved motifs and genomic localization among gene family members. A total of eight *ZmMET* genes were identified in maize. Moreover, we also isolated the cDNA sequences with high homology to genes encoding different DNA methyltransferase of these identified genes, and their expression patterns were analyzed by several approaches. The expression profiles of maize DNA methyltransferase genes under two different stress treatments were also investigated in order to gain insight into the understanding of the regulatory mechanisms of differential gene expression observed between the induced conditions and the control conditions.

Materials and methods

In silico identification of *ZmMET* genes, phylogenetic analysis and chromosomal localization

The C5-MTase proteins encoded in the entire maize genome were identified using search tools as described previously (Qian et al. 2011, 2014). Initially, a local BLAST database was constructed by the DNA TOOLS software (version 6.0; <http://www.dnatoools.com/>) and the HMM analysis was performed to extract an HMM profile for DNA methylase domain (PF00145) from the protein family (Pfam) database (<http://pfam.sanger.ac.uk/>). Second, the HMM profile was

used as a standard sequence to isolate all possible homologs in maize by BLASTp searches (P value = 0.001). Subsequently, the Pfam and SMART (<http://smart.embl-heidelberg.de/>) were used to determine each predicted protein as a member of the *ZmMET* gene family. The newly identified genes were designated on the basis of their phylogenetic relatedness to other members of the same gene family in *Arabidopsis* and rice. Basic information regarding the number of amino acids, clone number, ORF length and physical and chemical parameters of these genes were presented in this study. The exon–intron structure of these genes was determined using GSDS (<http://gsds.cbi.pku.edu.cn/>) by performing the comparison of the genomic sequences and their predicted coding sequences (CDS). The conserved motif divergence among *ZmMET* genes in maize was assessed by a complete amino acid online sequence analysis using MEME (multiple expectation maximization for motif elicitation) (Bailey and Elkan 1995). Moreover, a phylogenetic analysis was performed with the MEGA v4.0 program (Kumar et al. 2004) by the neighbor-joining method (Saitou and Nei 1987) and 1,000 bootstrap replicates were performed after aligning all protein sequences using Clustal-W (Thompson et al. 1994). Finally, the chromosome location image of all maize genes was generated by MapInspect (http://www.plantbreeding.wur.nl/UK/software_mapinspect.html) according to their starting positions in maize chromosomes. The gene duplication events of *ZmMETs* in maize were also investigated as described previously (Yang et al. 2008; Gu et al. 2002).

Plant materials and stress treatment

Maize (*Zea mays* L. inbred line B73) plants were grown in a growth chamber at 28 °C with a photoperiod of 15-h light and 9-h dark and a relative humidity of 60 %. Three-week-old seedlings were subjected to two abiotic stress treatments. Drought stress was induced by 15 % PEG-6000 (polyethylene glycerol), and seedling leaves were sampled at 24 h after the treatment. Seedling roots were submerged in 0.15 M NaCl solution for salt stress, and seedling leaves were sampled at 8 h after the treatment. The controls were treated with fresh water (Qian et al. 2011, 2014).

Expression patterns analyses of *ZmMET* genes

To confirm the expression patterns of the *ZmMET* genes, several approaches were performed in this study. Initially, the analysis of *ZmMETs* expression profiles was accomplished by searching the maize dbEST database from NCBI and finding expression information provided at the Web sites. Parameters of searching were as described previously (Qian et al. 2011, 2014). In addition to the maize EST database, maize expression data was also from the

PlantGDB and MAGI databases including EST, cDNA and PUTs (PlantGDB unique transcripts). In addition, the microarray data were analyzed by consulting with the transcriptome data published by Sekhon et al. (2011), which was publically available at MaizeGDB (<http://www.maizegdb.org>). Hierarchical clustering was performed with Cluster 3.0 and Treeview software (<http://bonsai.hgc.jp/mdehoon/software/cluster/>) based on the Pearson coefficients with average linkage to analyze gene expression patterns in multiple different tissues or organs, respectively. Finally, a semi-quantitative expression profile was also analyzed in the leaves by two different abiotic stress treatments to reveal the impact of exogenous factors on expression of DNA methyltransferase genes in maize referring to the methods as described previously (Qian et al. 2011, 2014).

Results

Identification of eight genes encoding ZmMET proteins in maize

HMM analysis resulted in the identification of eight genes encoding ZmMET proteins in the maize genome. Eight loci were confirmed as *ZmMET* genes in the maize genome on the basis of HMM analysis of the conserved catalytic C-terminal domain of C-5 cytosine-specific DNA methylase from the putative polypeptide sequence. The length of the *ZmMET* open reading frames (ORFs) varied from 1,074 bp for ZmMET4 to 4,611 bp for ZmMET1, with the respective coding potential of 357 and 1,536 amino acids (Table 1). The presence of conserved domains and motifs was verified using SMART analysis and NCBI databases, according to the characteristic of the plant C5-MTase proteins from DNA methyltransferase family, especially some model plants including *Arabidopsis* and rice. This result revealed that the ubiquitous presence of the conserved catalytic C-terminal domain with conserved motifs in all C5-MTase proteins from four subfamilies in maize (Fig. 1). However, the N-terminal domains of these C5-MTase proteins showed diverse structural features between different members of this family (Fig. 1). Similar to their counterparts in *Arabidopsis*, ZmMET1a or ZmMET1b contains two BAH domains in the N-terminal region of ZmMET1, while a single BAH domain is present in the ZmMET2a or ZmMET2b, as opposed to MET1 proteins, suggesting similar functions for these two C5-MTase families. Moreover, ZmMET2a or ZmMET2b is distinguished by the insertion of a chromodomain amino acid motif (CHR) between the conserved motifs II and IV. For ZmMET3a–3c, each contains two UBA domains in the N-terminal region of it, as opposed to DRM1/DRM2

proteins, which generally contain three UBA domains. Like other orthologues in *Arabidopsis*, these ZmMET3 proteins demonstrate circular permutation of their catalytic domain motifs, such that motifs VI–X precede motifs I–V. ZmMET4 shares a great resemblance to those of bacterial C5-MTases, with the lacking N terminus present in the other C5-MTases.

Sequence alignment, phylogenetic and structural analysis of *ZmMET* genes

The newly identified putative proteins were aligned with the known C5-MTase sequences in *A. thaliana*. Sequence alignment indicated that all ZmMET proteins contained six of the ten conserved motifs found in DNA methyltransferases (Fig. S1–3). Among them, the conserved catalytic motifs I and X are involved in the AdoMet binding, and a proline–cysteine doublet present in conserved motif IV has been identified as the functional catalytic site in cytosine-5 methyltransferases. Motif VI provides a glutamic acid that plays an important role in the target cytosine binding. Motif VIII is suggested to make nonspecific contacts with cytosine, which contribute to the neutralization of the negative charge of the DNA backbone. Motif IX is involved in the organization of the target recognition domain (TRD).

To determine the phylogenetic relationships between the plant C5-MTase genes, and assess the evolutionary history of these gene families among maize, rice and *Arabidopsis*, the full length protein sequences from these plants were used to construct an unrooted neighbor-joining phylogenetic tree (Fig. 2). The unrooted phylogenetic tree, generated from aligned full-length protein sequences of all eight ZmMETs, 10 OsMETs and 11 AtMETs, grouped maize, rice and *Arabidopsis* C5-MTase genes into four subfamilies, Dnmt1-like/MET, Dnmt2-like, CMT and Dnmt3-like/DRM, with well-supported bootstrap values. ZmMET1a and ZmMET1b were grouped into the first subfamily, representing the Dnmt1-like/MET class. These two maize genes were clustered with OsMET1 on the basis of high sequence similarity to OsMET1-1 (DMT702) and OsMET1-2 (DMT707). Similar to the Dnmt2-like class in rice and *Arabidopsis* C5-MTase gene families, there was only one protein found in the maize Dnmt2-like class, which was designated ZmMET4 and shared greater similarity with AtMET4 (DMT11) and OsMET4 (DMT705), respectively. ZmMET2a and ZmMET2b were grouped into the CMT subfamily, representing a class of chromomethylases that was found only in plants. The Dnmt3-like/DRM subfamily contains three maize C5-MTase genes, ZmMET3a–ZmMET3c, representing de novo methyltransferase that transfers methyl group to unmethylated DNA templates. Based on sequence comparisons, ZmMET3a and ZmMET3b shared higher

Table 1 Basic information of DNA methyltransferase genes identified in maize

Serial no.	Gene name	Accession number	Genome location		ORF length (bp)	Protein			Chr.	Type
		Ensembl transcript	Clone number	Coordinates (5'–3')		Length (aa)	Mol.wt. (Da)	PI		
1	<i>ZmMET1a</i>	GRM2M2G333916_P01	AC211651.4	11012377..11018774	4,611	1,536	172927.0	5.4	7	MET
2	<i>ZmMET1b</i>	GRM2M2G334041_P01	AC211651.4	11031086..11037097	4,611	1,536	172884.0	5.40	7	MET
3	<i>ZmMET2a</i>	GRMZM2G025592_P01	AC204716.3	141826364..14183411	2,745	914	101533.4	5.23	10	CMT
4	<i>ZmMET2b</i>	GRMZM2G005310_P02	AC190685.3	10250384..10260014	2,748	915	101610.8	5.44	2	CMT
5	<i>ZmMET3a</i>	GRMZM2G137366_P01	AC186330.4	2910056..2915015	1,830	609	67,696.5	5.13	1	DRM
6	<i>ZmMET3b</i>	GRMZM2G092497_P06	AC191798.3	150137673..15014005	1,812	603	66,821.5	5.15	9	DRM
7	<i>ZmMET3c</i>	GRMZM2G065599_P01	AC206320.3	128612399..12861458	1,383	460	52,326.3	6.58	8	DRM
8	<i>ZmMET4</i>	GRMZM2G157589_P01	AC198121.4	141275856..141280632	1,074	357	40,538.3	5.83	8	Dnmt2

The information of maize DNA methyltransferase gene family, including accession number, chromosomal location and ORF length, were retrieved from the B73 maize sequencing database (<http://www.maizesequence.org/index.html>)

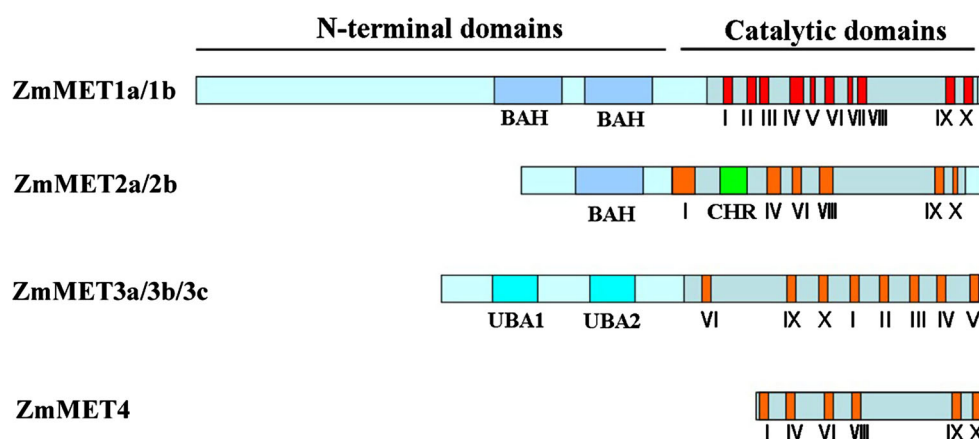


Fig. 1 Schematic structures of maize DMT proteins. Schematic representation of conserved domains and motifs identified among each subfamily of maize DMT proteins. Red rectangles represent the conserved motifs in the catalytic domains, A green rectangle

indicates the conserved CHR domain in the catalytic of MTase and the BAH and UBA conserved domains of N-terminal regulatory region are represented by purple and blue–green rectangles, respectively (color figure online)

similarity with two *Arabidopsis* genes (DMT7, DMT9) and three rice genes (DMT706, DMT708, DMT709), while *ZmMET3c* exhibited the higher similarity to *Arabidopsis* gene DMT10 and rice gene DMT710. Subsequently, an exon–intron structure analysis was performed to support the phylogeny reconstruction. The schematic structures revealed that each coding sequence of *ZmMET* gene is disrupted by one or more exons. The *ZmMET* genes within the same groups of the phylogenetic tree all showed similar exon–intron structures (Fig. 3).

Analysis of conserved motifs in C5-MTase proteins

The results from the preliminary Pfam analyses of the entire predicted proteins were applied to the MEME motif search tool. This analysis was used to further identify conserved motifs in the corresponding conserved domains of all eight C5-MTase proteins encoded by this gene family

from maize. The search was performed for all predicted proteins and 15 conserved motifs were identified (Fig. 4; Table 2). Moreover, SMART and Pfam were further used to annotate the motifs identified by MEME. The majority of the motifs were well conserved in the Dnmt1-like/MET and CMT subfamilies, but the Dnmt2-like and Dnmt3-like/DRM subfamilies lost more than half of these motifs identified in this study. Individual and sequential manual motif analyses using SMART and Pfam programs were performed to further elucidate the major functional roles of these conserved motifs in this family. This result shows the annotated conserved motifs 7, 2, 5, 3, 4, 10, 8 and 1 specified for the DNA methylase domain in the C-terminal of the DMT protein; motif 11 for the BAH domain and motif 15 for the CHR domain. Interestingly, the Dnmt-like/MET and CMT subfamilies share a common motif (motif 15), which exists in a different location, but it might remain a key role in maintenance methylation in plants.

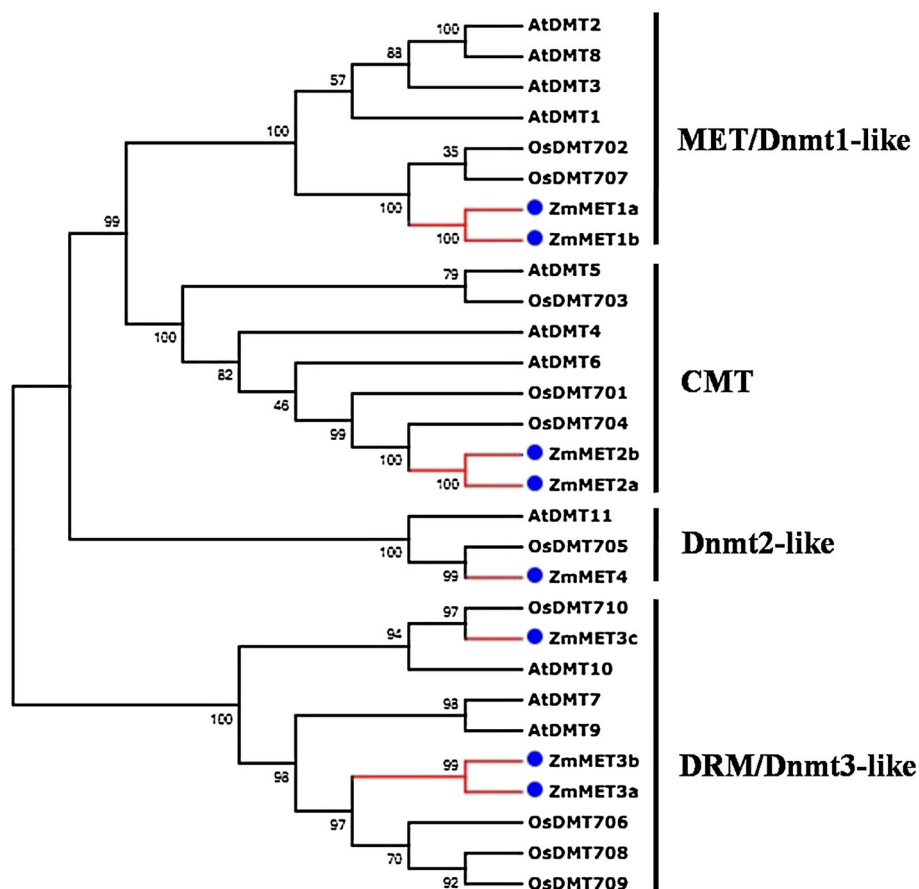


Fig. 2 Phylogenetic relationship among DMT proteins of maize, rice and *Arabidopsis*. Unrooted neighbor-joining (NJ) phylogenetic tree of maize, rice and *Arabidopsis* DMT proteins with bootstrap values shown for each clade. The maize ZmMETs have been highlighted for each group. Four clades are marked: Dnmt1-like/MET, Dnmt2-like, CMT and Dnmt3-like/DRM. Protein sequences were downloaded from National Center for Biotechnology Information (NCBI). Accession numbers and abbreviations are as follows:

DMT1 (At5g49160), DMT2 (At4g14140), DMT3 (At4g13610), DMT4 (At1g80740), DMT5 (At4g19020), DMT6 (At1g69770), DMT7 (At5g14620), DMT8 (At4g08990), DMT9 (At5g15380), DMT10 (At3g17310), DMT11 (At5g25480); DMT701 (Os03g12570), DMT702 (Os03g58400), DMT703 (Os05g13790), DMT704 (Os10g01570), DMT705 (Os01g42630), DMT706 (Os03g02010), DMT707 (Os07g08500), DMT708 (Os12g01800), DMT709 (Os11g01810), DMT710 (Os05g04330)

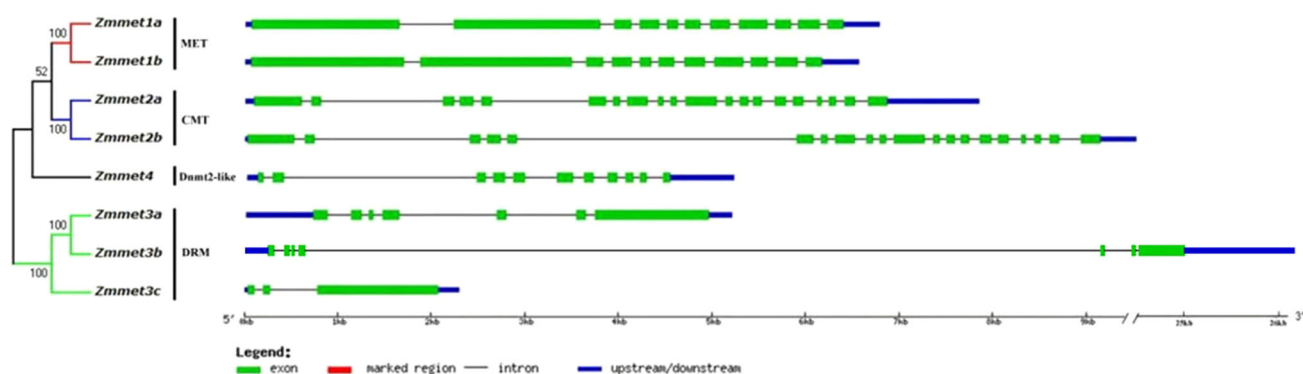


Fig. 3 Phylogenetic relationship and gene structure of the eight predicted maize DMT proteins. **a** The unrooted tree was constructed with MEGA4.0 program using the full-length amino acid sequences of the eight maize DMT proteins. The bootstrap values are indicated at the branches in black numbers, and the proteins were named

according to their gene codes. **b** Exons and introns are indicated by white green boxes and single lines, respectively. Thick blue lines represent the untranslated regions (UTRs). The length of each DMT gene can be estimated using the scale at the bottom (color figure online)

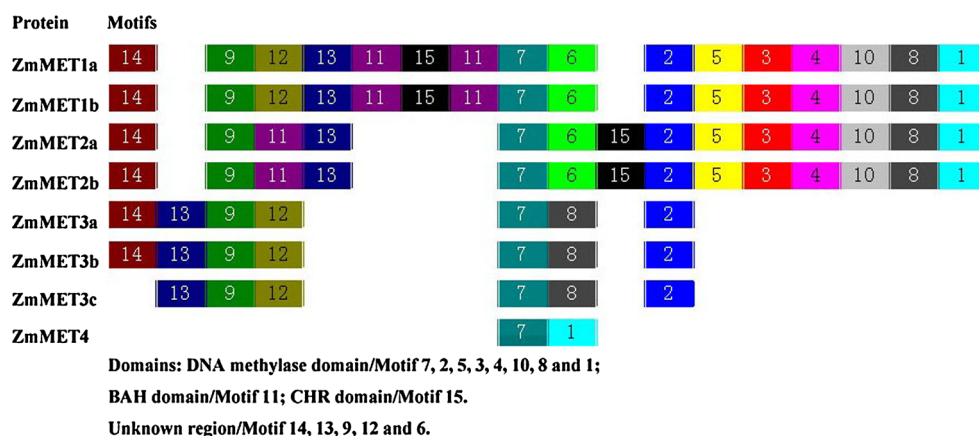


Fig. 4 Distribution of conserved motifs in maize DMT proteins identified using the MEME search tool. Schematic representation of motifs identified in maize DMT proteins using MEME motif search tool for each group. Each motif is represented by a number in the box.

Box length does not correspond to length of motif. Order of the motifs corresponds to position of motifs in individual protein sequence. For motif detail refer to Table 2

Table 2 Detailed information on the 15 *ZmMET* protein conserved motifs

Motif	Width	Conserved amino acid sequences
1	50	HPTQDRIITIRECARLQGFDPDYRFFGPIQNKHRQIGNAVPPPLAYALGY
2	29	NYLPLPGEVDFICGGPPCQGFSGMNRHHQ
3	41	YQVRFGMMVAGCYGLPQFRMRVFIWGAAPGMMLPDWPLPMH
4	50	FKAITLGDTIGDLPKVGHNQPNDTMEYGGPKTWFQKYIRGNMMDMNDWI
5	50	DVQCKQMLTFMDFVEYFKPKYFLMENVDRFLKFNDGYTFKYALACLLEM
6	50	FTKWAIDYNEFACAFNYNHPQTTVVFNDNCDVFLKAIMDWCGDCDDCVHT
7	21	TVLDIYSGCGGMSTGLHQLGI
8	29	MSFEHNQWNRPYGRLWWDQTQPTTVTRPQ
9	29	PPFGMTICEYIPRTTRWWPSWDPCHQFTC
10	32	MNNDYIRCQHIPPCKPGCDWHDLPGVKVGANN
11	29	GHKHDPRRVYYSKDNDNPLDCIIGKCKI
12	41	APLIPDVMEFDKGFPHFTRGITRTERYRVLHGQFQLQQA
13	28	PKAMWHTTTRFYDIWPEFYHHFFCAS
14	40	GCRGPIDFICHEPMPHPMAMFEIHDHWPTRYDRSMPAQ
15	50	GQPMVYFCYSGYWPEKGAYFKLQWDGMGPGEGTWEPDNIIEPCDKIKIF

These analyses clearly demonstrated that these motifs were well conserved and shared major functional roles in these proteins.

Chromosomal localization and gene duplication

The physical locations of *ZmMET* genes in maize were investigated by analysis of genomic distribution on chromosomes. The eight genes were found to be distributed evenly across six of all the ten chromosomes in the maize genome (Fig. 5). Each chromosome was located with one or two genes almost distributed in the end of the chromosome, respectively.

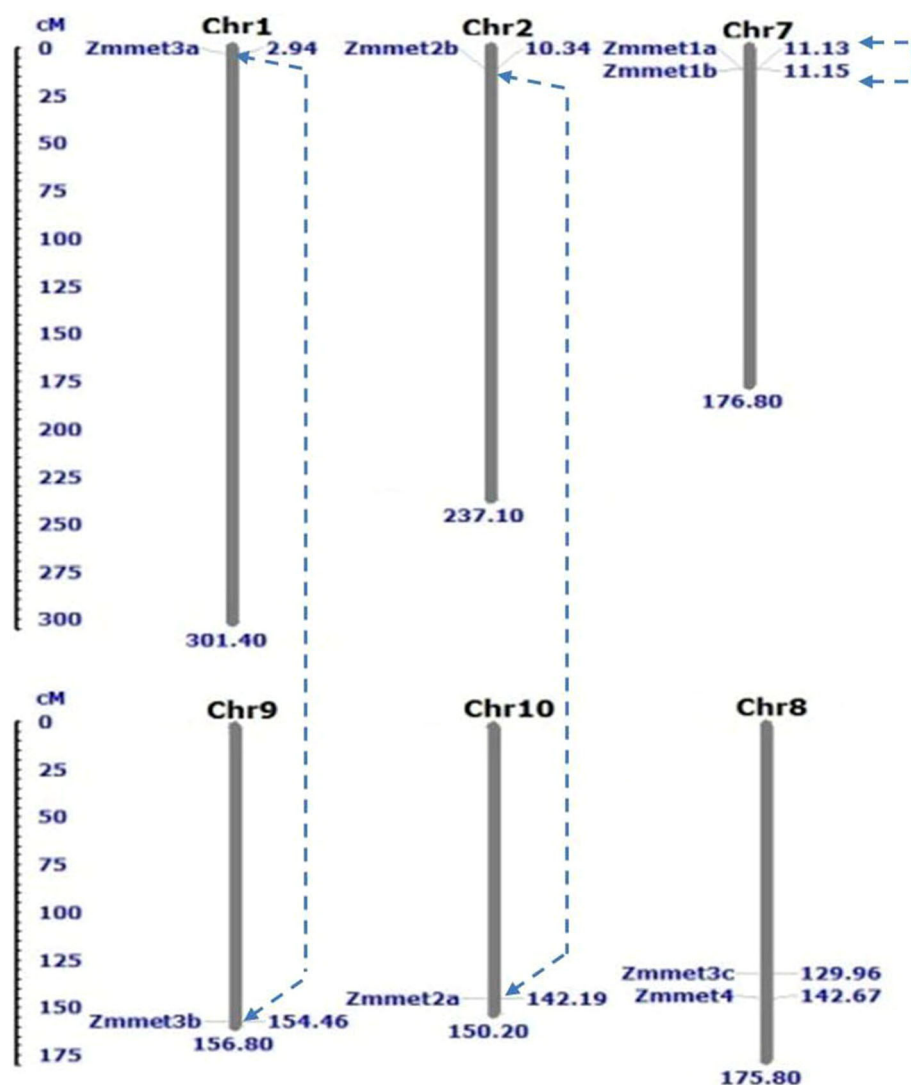
To elucidate the expanded mechanism of the *ZmMET* gene family, gene duplication events including tandem and segmental duplications, which are thought to have occurred

during the process of evolution, were investigated in maize. Based on the phylogenetic analysis and the chromosomal distribution of the *ZmMET* genes, two gene pairs (*ZmMET2a/ZmMET2b* and *ZmMET3a/ZmMET3b*) were identified to be involved in the segmental duplication events (Fig. 5), which exist in the two subfamilies, CMT and Dnmt3-like/DRM, respectively. Furthermore, one tandem duplicated gene pair (*ZmMET1a/ZmMET1b*) was found in the Dnmt1-like/MET subfamily (Fig. 5).

Digital expression patterns analysis: EST and microarray expression profiles

Gene expression patterns are usually related to identification of gene function. In this study, two digital analyses were performed to reveal the diversity of maize *ZmMET*

Fig. 5 Chromosomal locations of maize DMT genes on all 10 chromosomes. Chromosome numbers are shown at the top of each vertical gray bar. The gene names on the left side of each chromosome correspond to the approximate locations of each DMT gene. The segmental duplication genes are connected by dashed lines. The scale on the left is in megabases



gene expression patterns. Initially, the digital expression profiles of the *ZmMET* genes were identified using corresponding EST databases with known *ZmMETs* coding sequences, resulted in the assignment of *ZmMETs* to 12 groups on the basis of tissue and organ types released by the maize FLCDNA project and other expression evidence was verified in MAGI and PlantGDB databases (Table 3). After integrating and analyzing all EST expression data, we found all *ZmMETs* were supported by expression evidence. Interestingly, most of these genes were shown to obtain expression evidence from only five of all twelve groups on the basis of tissue and organ types. Among them, there were three *ZmMET* genes that were revealed to express in only mixed tissues and to have no tissue-specific expression pattern. The others (five *ZmMET* genes) were shown to specifically express in one or more tissue and organ types. Furthermore, the expression pattern of three *ZmMET* duplicated gene pairs was also investigated, but two

paralogs of each gene pair exhibited dissimilar expression patterns. For example, *ZmMET1a* was detected only in mixed tissues; however, its paralogue gene *ZmMET1b* appeared to have tissue-specific expression pattern in embryo. Moreover, a comprehensive expression analysis was performed based on whole-genome coverage according to the microarray data released by Sekhon et al. (2011). By clustering expression profiles, the majority of the *ZmMET* genes showed distinct tissue-specific expression patterns across the nine tissues or organs investigated. For instance, three genes (*ZmMET3a*, *ZmMET3b* and *ZmMET3c*) showed higher transcript abundances in the seedlings, whereas other two genes (*ZmMET2a* and *ZmMET2b*) had a lower expression in the stems (Fig. 6). The results revealed that these genes belong to even the same subfamily exhibited more concentrated expressions in the seedlings or stems than the other tissues or organs, respectively.

Table 3 Expression analysis of *ZmMET* genes in silico

Gene	Tissue and organ type (NCBI)										Number of ESTs in dbEST			MAGI	PlantGDB		PUTs
	Silks	Husks	Ear	Tassel	Shoot tips	Leaf	Root tips	Seeding	Embryo	Endosperm	Seed	Mixed	EST		cDNA		
<i>ZmMET1a</i>												+	7	+	+	+	
<i>ZmMET1b</i>									+			+	7	+	+	+	
<i>ZmMET2a</i>										+		+	10	+	+	+	
<i>ZmMET2b</i>									+	+		+	7	+	+	+	
<i>ZmMET3a</i>					+					+			2	+	+	+	
<i>ZmMET3b</i>												+	1	+	+	+	
<i>ZmMET3c</i>										+		+	1	+	+	+	
<i>ZmMET4</i>											+	+	6	+	+	+	
+ expressed; blank not expressed																	

Expression analysis of *ZmMET* gene family under stress treatment

To further confirm these predicted genes and understand their expression profiles under various stress conditions, two abiotic stress treatments (PEG for drought stress and NaCl for salt stress) were investigated. Semi-quantitative RT-PCR analyses on RNA isolated from maize leaves were performed (Fig. 7). The results revealed that these genes were differentially expressed in the leaves under either normal condition (control) or stress conditions (15 % PEG or 0.15 M NaCl treatment). On the basis of the brightness of the bands, four of the expressed genes (*ZmMET1a*, *ZmMET1b*, *ZmMET2a* and *ZmMET2b*) showed decreased expression levels in maize leaves by PEG or NaCl treatment, whereas other four genes including *ZmMET3a*, *ZmMET3b*, *ZmMET3c* and *ZmMET4* showed no obvious difference in the leaves of maize under these two stress conditions. These results demonstrated that these predicted genes of the MET and CMT subfamilies exhibit different expression levels in two stress treatments in this study.

Discussion

In plants, DNA methylation is an important epigenetic mark involved in diverse biological processes including regulation of gene expression during development and chromatin organization (Finnegan et al. 1996, 1998). As one of the main ways of genome control, DNA methylation has been implicated in repressing the transcription of invading and mobile DNA elements, such as transgenes, viruses, transposons, and retroelements in plants (Kidwell and Lisch 2001). In plants, DNA methylation is conserved and mainly occurs in the contexts of CG, CHG, and CHH (H = A, C, or T), which mainly depends on the presence of the plant cytosine-5 DNA methyltransferases (C5-MTases). Recent studies have demonstrated that the plant C5-MTases that have been identified so far are classified into four main subfamilies based on evolutionary relatedness and their linear domain arrangement. The plant different C5-MTase protein interacting residues are dispersed throughout the entire C5-MTase proteins. However, with diverse functions, the evolutionary relationship of C5-MTase proteins family has been poorly studied in maize.

In this present study, a comprehensive set of eight nonredundant C5-MTase proteins was identified and characterized from the latest version of the maize (B73) genome. The result indicated that there were fewer C5-MTase-encoding repeat genes in maize (8) compared with rice (10) and *Arabidopsis* (11), despite the maize genome (2300Mbp) being approximately several-fold larger than that of rice (389Mbp) and *Arabidopsis* (125Mbp). As an

Fig. 6 Clustering of expression profile of eight *ZmMET* genes. The color scale representing log₂ signal values is shown on the left, Green represents a low level and red indicates a high level of transcript abundance (color figure online). The different organisms are noted on the top of each lane. Cluster dendrogram is shown on the left

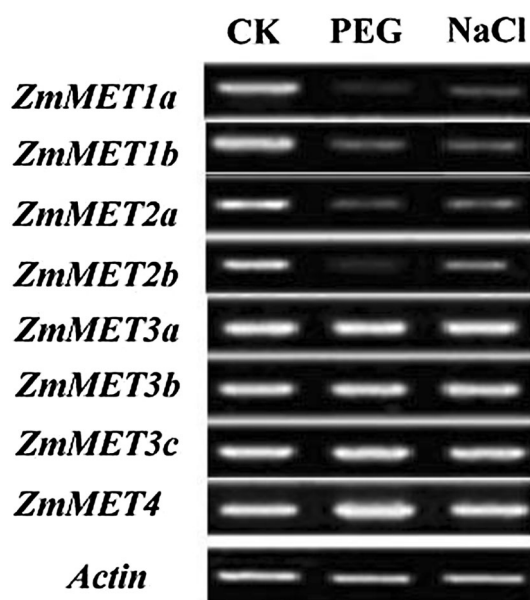
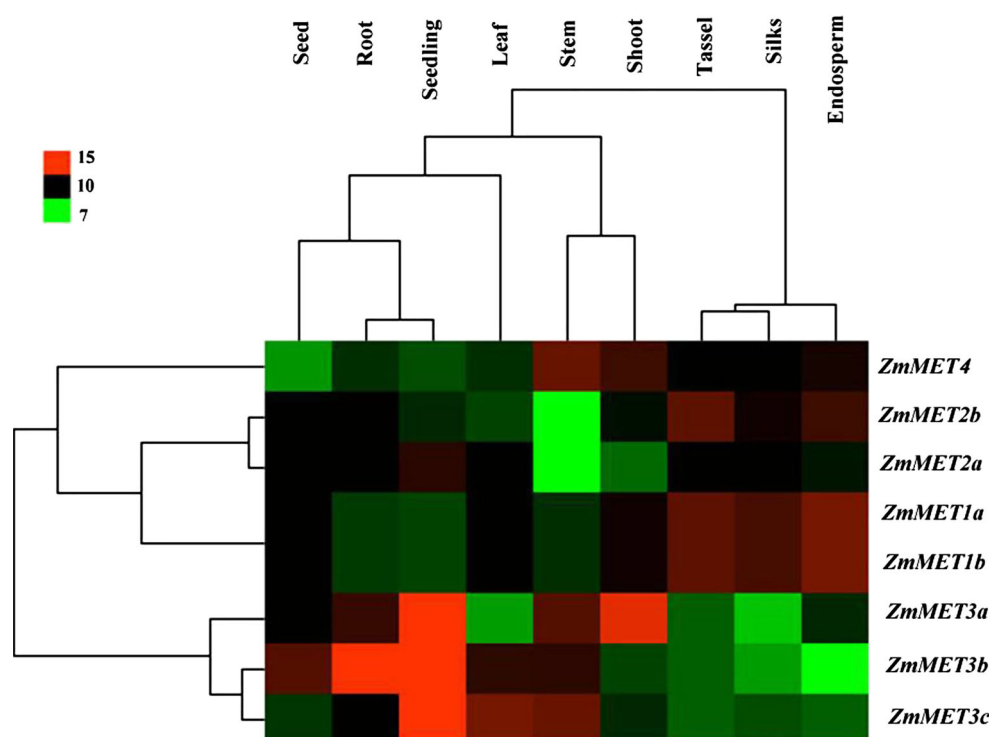


Fig. 7 Semi-quantitative RT-PCR analysis of *ZmMET* genes under stress treatments. Total RNA was extracted from the 21-day-old seedlings germinated on wet filter paper soaked with distilled water and grown in a growth chamber. Either fresh water (1), 15 % PEG (2) or .15 M NaCl (3) was applied 24 h before harvest. An amplified maize actin gene was used as an internal control

interesting result, we have inferred that both gene duplication and gene loss may play an important role in the course of evolution of this gene family. The evolution of C5-MTase-encoding gene family may be complicated and

experienced a large gene loss during maize genome duplication. Furthermore, phylogenetic analysis provided insights into the evolution of gene family members and gene multiplicity in maize. In our investigation of conserved C5-MTase proteins, we found that maize C5-MTase-encoding genes were classified into four distinct classes, which is also similar to those present in *Arabidopsis* and rice. This further demonstrated that maize shares a close relationship with *Arabidopsis* and rice during the evolutionary process. Furthermore, phylogenetic analysis of C5-MTase-encoding genes in maize, rice and *Arabidopsis* indicated that *ZmMETs* are more closely allied with *OsDMTs* than *AtDMTs*, consistent with the evolutionary relationships among maize, rice and *Arabidopsis*. The fact that all four classes are identified in maize, rice and *Arabidopsis* genes implies that the C5-MTase-encoding genes originated prior to the divergence of monocots and dicots. It is suggested that the expansion of these C5-MTase-encoding genes following divergence of monocots and dicots.

According to previous reports, in *Arabidopsis*, increasing genetic and biochemical evidence indicates that the function of members from the Dnmt1-like/MET and CMT classes are demonstrated to be responsible for the maintenance of GpG and CpNpG methylation, respectively (Cao et al. 2003; Bartee and Bender 2001), whereas the domains-rearranged methyltransferases (DRMs) are involved in the principal de novo methylation. However, until now, the role of the Dnmt2-like class still remains

largely mysterious and unelucidated since they lack the N terminus present in the other C5-MTases in plants, which shares a great resemblance to the structural features of bacterial C5-MTases. These results further revealed that diverse mechanisms could be involved in shaping the function and regulation of these C5-MTase proteins. Furthermore, the characterization of conserved domains in maize C5-MTase proteins also contributes to elucidate the mechanism diversity in shaping the function and regulation of these C5-MTase proteins. The results of domain analysis of the plant C5-MTase proteins indicated that the catalytic C-terminal domain remained strictly conserved whereas the N-terminal domain shared relatively variable feature among different C5-MTase protein subfamilies during evolutionary process of this family in plants. A possible excuse is that the catalytic C-terminal domain may take the conserved functions in plants, which has been demonstrated to be responsible for mainly accomplishing the binding of C5-MTases to the DNA bases and the cofactor AdoMet (Jeltsch 2006). However, the N-terminal domain may be related with recognition of a specific DNA sequence through several N-terminal special domains or motifs of the C5-MTases in plants. Additionally, the phylogenetic reconstruction was also supported by analyzing gene structure of the *ZmMET* genes, which exhibited highly conserved in evolution. Closed related *ZmMET* genes subfamily members in the phylogenetic tree shared similar exon–intron structure. Also, the conserved evolution of *ZmMET* genes was reflected by the domains defined by motif distribution and gene duplication events investigated in this study.

In addition, gene expression research information can usually profit from obtaining gene expression in silico data from EST databases and microarray data from the transcriptome dataset publically available at MaizeGDB, which plays an increasingly vital role in facilitating the identification of gene function and future functional genomics studies in plant growth and development (Qian et al. 2011, 2014). In this study, several approaches were used for maize C5-MTase-encoding gene expression patterns in different tissues or organs. By EST expression data mining and expression profiles clustering, these newly identified genes exhibited distinct temporal and spatial expression patterns in different tissues or organs. This result further revealed that the expression of these genes may occur under specific environment or very specific to organ or developmental stage or genotypes of maize. Furthermore, three duplicated *ZmMET* gene pairs also exhibited diverse expression patterns between two members, which revealed that functional diversification of the surviving duplicated genes is a major feature of the long-term evolution (Blanc and Wolfe 2004). Moreover, to further reveal how these genes reacts to stress

responses, the expression profiles of maize under two different stress treatments were investigated by semi-quantitative RT-PCR in this study. The result displayed that four predicted genes of the MET and CMT subfamilies exhibited different expression levels under two different abiotic stress treatments, whereas the other subfamilies (DRM and Dnmt2-like class) remained non-responsive to the two abiotic stress conditions in this study. One possibility is that different regulation pathways might exist in controlling the establishment and maintenance of DNA methylation in plants, especially under specific stress environment. Our results suggest that the *MET* and *CMT* genes might play important roles in for plant DNA methylation regulation by showing strong response to different abiotic stress conditions to survive under some specific stress environment in plant development. In contrast, while the *DRM* and *Dnmt2-like* genes showed no obvious difference in the leaves of maize under these two stress conditions in this study, they still remain a high possibility to contribute to maize DNA methylation regulation by only expressing under specific conditions or in specific tissues other than seedling leaves, and these remain to be further confirmed experimentally.

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Conflict of interest The authors declare that they have no conflict of interests.

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